

	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi}$					
$\mathcal{K}_{0^+}^{\#1} \dagger$	$\frac{1}{2}(-\Upsilon_1 k^2 - 3 \binom{2}{K_3})$	0					
$\mathcal{K}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	$\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#1}{}_{\alpha}$	$\mathcal{K}_{1^+}^{\#2}{}_{\alpha}$	
	$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{\Upsilon_2 k^2}{2}$	$-\frac{\Upsilon_3 k^2}{2\sqrt{2}}$	0	0		
	$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$-\frac{\Upsilon_3 k^2}{2\sqrt{2}}$	$-\frac{\Upsilon_4 k^2}{4}$	0	0		
	$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha}$	0	0	$\frac{1}{2}(\Upsilon_5 k^2 - 2 \binom{2}{K_3})$	$\frac{-\Upsilon_5 k^2 + 2 \binom{2}{K_3}}{2\sqrt{2}}$		
	$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha}$	0	0	$\frac{-\Upsilon_5 k^2 + 2 \binom{2}{K_3}}{2\sqrt{2}}$	$\frac{1}{4}(\Upsilon_5 k^2 - 2 \binom{2}{K_3})$	$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$
					$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{\Upsilon_6 k^2}{2}$	0
					$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{3 k^2 \binom{4}{K_{15}}}{2}$

[illegible]

$$\begin{aligned}
 & \text{Abbreviations used in matrices} \\
 & Y_1 == 3 \overset{(4)}{\kappa}_1 - 3 \overset{(4)}{\kappa}_{15} + 3 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6 \& Y_2 == \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6 \& \\
 & Y_3 == 2 \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_4 \& Y_4 == 5 \overset{(4)}{\kappa}_{15} - 2 \left(\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 \right) \& Y_5 == 3 \overset{(4)}{\kappa}_{15} - 2 \overset{(4)}{\kappa}_2 \& \\
 & Y_6 == 3 \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_6 \& \text{Det}(0^+) == \frac{1}{2} \left(-k^2 \left(3 \overset{(4)}{\kappa}_1 - 3 \overset{(4)}{\kappa}_{15} + 3 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6 \right) - 3 \overset{(2)}{\kappa}_3 \right) \& \\
 & \text{Det}(1^+) == -\frac{1}{8} k^4 \left(9 \overset{(4)}{\kappa}_{15}^2 + 2 \overset{(4)}{\kappa}_3^2 + 2 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4^2 - 2 \left(\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 \right) \overset{(4)}{\kappa}_6 + \overset{(4)}{\kappa}_{15} \left(-7 \overset{(4)}{\kappa}_3 - 6 \overset{(4)}{\kappa}_4 + 5 \overset{(4)}{\kappa}_6 \right) \right) \& \\
 & \text{Det}(2^+) == \frac{1}{2} k^2 \left(3 \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_6 \right)
 \end{aligned}$$

$$\begin{aligned} & \text{Lagrangian} \\ & \frac{(2)}{\kappa_3} \mathcal{K}^\alpha{}_\alpha{}^\beta{}_\beta \mathcal{K}^\chi{}_\chi{}^\gamma{}_\gamma + \frac{(4)}{\kappa_1} \partial_\beta \mathcal{K}^\delta{}_\chi{}^\gamma{}_\gamma \partial^\chi \mathcal{K}^\alpha{}_\alpha{}^\beta{}_\beta + \frac{(4)}{\kappa_2} \partial_\chi \mathcal{K}^\delta{}_\beta{}^\gamma{}_\gamma \partial^\chi \mathcal{K}^\alpha{}_\alpha{}^\beta{}_\beta + \frac{(4)}{\kappa_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi}{}_\gamma \partial_\delta \mathcal{K}^\delta{}_\alpha{}^\chi{}_\chi + \\ & \frac{(4)}{\kappa_4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi}{}_\gamma \partial_\delta \mathcal{K}^\delta{}_\beta{}^\chi{}_\chi + \frac{(4)}{\kappa_6} \partial_\beta \mathcal{K}^{\alpha\beta\chi}{}_\gamma \partial_\delta \mathcal{K}^\delta{}_\chi{}^\alpha{}_\alpha - 3 \frac{(4)}{\kappa_{15}} \partial^\chi \mathcal{K}^\alpha{}_\alpha{}^\beta{}_\beta \partial_\delta \mathcal{K}^\delta{}_\chi{}^\beta{}_\beta - \frac{9}{4} \frac{(4)}{\kappa_{15}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi}{}_\gamma \partial_\delta \mathcal{K}^\delta{}_\beta{}^\chi{}_\chi + \\ & \frac{1}{2} \frac{(4)}{\kappa_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi}{}_\gamma \partial_\delta \mathcal{K}^\delta{}_\beta{}^\chi{}_\chi + \frac{1}{2} \frac{(4)}{\kappa_4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi}{}_\gamma \partial_\delta \mathcal{K}^\delta{}_\beta{}^\chi{}_\chi + \frac{(4)}{\kappa_{15}} \partial_\delta \mathcal{K}^\alpha{}_\alpha{}^\beta{}_\beta \partial^\delta \mathcal{K}^{\alpha\beta\chi}{}_\gamma + \frac{(4)}{\kappa_{15}} \partial_\delta \mathcal{K}^\delta{}_\beta{}^\chi{}_\chi \partial^\delta \mathcal{K}^{\alpha\beta\chi}{}_\gamma \end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_0^{\#1\alpha\beta\chi} = 0$	1	$\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} =$ $\partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}_1^{\#1\alpha} + 2 \mathcal{J}_1^{\#2\alpha} = 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}^{\beta\chi}_{\beta\chi} + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}_{\beta\alpha} = 3 \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$
Total # constraint(s):	4	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$-\frac{-6 \kappa_{15}^{(4)} \kappa_3^{(4)} \kappa_6^{(4)}}{\kappa_{15}^{(4)} (-3 \kappa_{15}^{(4)} \kappa_3^{(4)} \kappa_6^{(4)})}$

	2	0	$-\frac{6 \overset{(4)}{\kappa} \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_6}{\overset{(4)}{\kappa}_{15} (-3 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6)}$
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	1	0	$ \begin{aligned} & -((12 \binom{(4)}{\kappa_{15}}^2 (2 \binom{(4)}{\kappa_3} - \binom{(4)}{\kappa_4} - 2 \binom{(4)}{\kappa_6}) - 4 \binom{(4)}{\kappa_{15}} (2 \binom{(4)}{\kappa_3} - \binom{(4)}{\kappa_4} - 2 \binom{(4)}{\kappa_6}) (\binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_6}) - \sqrt{((-3 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_6})^2} \\ & (324 \binom{(4)}{\kappa_{15}}^4 + (2 \binom{(4)}{\kappa_3}^2 + \binom{(4)}{\kappa_4} (\binom{(4)}{\kappa_4} - 2 \binom{(4)}{\kappa_6}) + 2 \binom{(4)}{\kappa_3} (\binom{(4)}{\kappa_4} - \binom{(4)}{\kappa_6}))^2 - 36 \binom{(4)}{\kappa_{15}}^3 (13 \binom{(4)}{\kappa_3} + 8 \binom{(4)}{\kappa_4} - 9 \binom{(4)}{\kappa_6}) + \\ & \binom{(4)}{\kappa_{15}}^2 (369 \binom{(4)}{\kappa_3}^2 + 152 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_4} + 132 \binom{(4)}{\kappa_4}^2 - 562 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_6} - 88 \binom{(4)}{\kappa_4} \binom{(4)}{\kappa_6} + 209 \binom{(4)}{\kappa_6}^2) - 2 \binom{(4)}{\kappa_{15}} (26 \binom{(4)}{\kappa_3}^3 + \\ & \binom{(4)}{\kappa_3}^2 (42 \binom{(4)}{\kappa_4} - 44 \binom{(4)}{\kappa_6}) + \binom{(4)}{\kappa_3} (29 \binom{(4)}{\kappa_4}^2 - 60 \binom{(4)}{\kappa_4} \binom{(4)}{\kappa_6} + 18 \binom{(4)}{\kappa_6}^2) + \binom{(4)}{\kappa_4} (8 \binom{(4)}{\kappa_4}^2 - 25 \binom{(4)}{\kappa_4} \binom{(4)}{\kappa_6} + 18 \binom{(4)}{\kappa_6}^2))))) / \\ & (\binom{(4)}{\kappa_{15}} (-3 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_6}) (9 \binom{(4)}{\kappa_{15}}^2 + 2 \binom{(4)}{\kappa_3}^2 + \binom{(4)}{\kappa_4} (\binom{(4)}{\kappa_4} - 2 \binom{(4)}{\kappa_6}) + 2 \binom{(4)}{\kappa_3} (\binom{(4)}{\kappa_4} - \binom{(4)}{\kappa_6}) + \\ & \binom{(4)}{\kappa_{15}} (-7 \binom{(4)}{\kappa_3} - 6 \binom{(4)}{\kappa_4} + 5 \binom{(4)}{\kappa_6})))) \end{aligned} $
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	1	0	$ \begin{aligned} & -((12 \kappa_{15}^2 (2 \kappa_3 - \kappa_4 - 2 \kappa_6) - 4 \kappa_{15}^2 (2 \kappa_3 - \kappa_4 - 2 \kappa_6) (\kappa_3 + \kappa_6) + \sqrt{((-3 \kappa_{15} + \kappa_3 + \kappa_6)^2 \\ & (324 \kappa_{15}^4 + (2 \kappa_3^2 + \kappa_4^2 (\kappa_4 - 2 \kappa_6) + 2 \kappa_3 (\kappa_4 - \kappa_6))^2 - 36 \kappa_{15}^3 (13 \kappa_3 + 8 \kappa_4 - 9 \kappa_6) + \\ & \kappa_{15}^2 (369 \kappa_3^2 + 152 \kappa_3 \kappa_4 + 132 \kappa_4^2 - 562 \kappa_3 \kappa_6 - 88 \kappa_4 \kappa_6 + 209 \kappa_6^2) - 2 \kappa_{15} (26 \kappa_3^3 + \\ & \kappa_3^2 (42 \kappa_4 - 44 \kappa_6) + \kappa_3 (29 \kappa_4^2 - 60 \kappa_4 \kappa_6 + 18 \kappa_6^2) + \kappa_4 (8 \kappa_4^2 - 25 \kappa_4 \kappa_6 + 18 \kappa_6^2))))) / \\ & (\kappa_{15} (-3 \kappa_{15} + \kappa_3 + \kappa_6) (9 \kappa_{15}^2 + 2 \kappa_3^2 + \kappa_4^2 (\kappa_4 - 2 \kappa_6) + 2 \kappa_3 (\kappa_4 - \kappa_6) + \\ & \kappa_{15} (-7 \kappa_3 - 6 \kappa_4 + 5 \kappa_6))) \end{aligned} $
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	2	0	$ \begin{aligned} & (81 \kappa_{15}^{(4)3} - 20 \kappa_3^{(4)3} - 20 \kappa_3^{(4)2} \kappa_4^{(4)} - 10 \kappa_3^{(4)} \kappa_4^{(4)2} - 18 \kappa_{15}^{(4)} (10 \kappa_3^{(4)} + 3 \kappa_4^{(4)} - 5 \kappa_6^{(4)}) - 10 \kappa_4^{(4)2} \kappa_6^{(4)} + 20 \kappa_3^{(4)} \kappa_6^{(4)} + \\ & 20 \kappa_4^{(4)} \kappa_6^{(4)} + \kappa_{15}^{(4)} (103 \kappa_3^{(4)} + 78 \kappa_3^{(4)} \kappa_4^{(4)} + 18 \kappa_4^{(4)2} - 40 \kappa_3^{(4)} \kappa_6^{(4)} + 6 \kappa_4^{(4)} \kappa_6^{(4)} - 71 \kappa_6^{(4)2}) + \sqrt{(183708 \kappa_{15}^{(4)6} - \\ & 1458 \kappa_{15}^{(4)5} (245 \kappa_3^{(4)} + 132 \kappa_4^{(4)} - 37 \kappa_6^{(4)}) + 100 (2 \kappa_3^{(4)2} + \kappa_4^{(4)} \kappa_4^{(4)} - 2 \kappa_6^{(4)}) + 2 \kappa_3^{(4)} (\kappa_4^{(4)} - \kappa_6^{(4)})^2 (\kappa_3^{(4)} + \kappa_6^{(4)})^2 + \\ & 243 \kappa_{15}^{(4)4} (1273 \kappa_3^{(4)2} + 330 \kappa_4^{(4)2} + 4 \kappa_3^{(4)} (309 \kappa_4^{(4)} - 97 \kappa_6^{(4)}) - 156 \kappa_4^{(4)} \kappa_6^{(4)} - 89 \kappa_6^{(4)2}) - \\ & 9 \kappa_{15}^{(4)3} (16558 \kappa_3^{(4)3} + 1836 \kappa_4^{(4)3} + 12 \kappa_3^{(4)2} (1827 \kappa_4^{(4)} - 545 \kappa_6^{(4)}) - 729 \kappa_4^{(4)2} \kappa_6^{(4)} - \\ & 2700 \kappa_4^{(4)} \kappa_6^{(4)2} + 1060 \kappa_6^{(4)3} + 3 \kappa_3^{(4)} (3645 \kappa_4^{(4)2} - 1800 \kappa_4^{(4)} \kappa_6^{(4)} - 1586 \kappa_6^{(4)2}) - \\ & 5 \kappa_{15}^{(4)} (\kappa_3^{(4)} + \kappa_6^{(4)}) (1256 \kappa_3^{(4)4} + 8 \kappa_3^{(4)3} (289 \kappa_4^{(4)} - 197 \kappa_6^{(4)}) + 2 \kappa_3^{(4)2} (995 \kappa_4^{(4)2} - 1400 \kappa_4^{(4)} \kappa_6^{(4)} - 16 \kappa_6^{(4)2}) + 2 \kappa_3^{(4)} (\\ & (417 \kappa_4^{(4)3} - 845 \kappa_4^{(4)2} \kappa_6^{(4)} + 68 \kappa_4^{(4)} \kappa_6^{(4)2} + 176 \kappa_6^{(4)3}) + \kappa_4^{(4)} (153 \kappa_4^{(4)3} - 390 \kappa_4^{(4)2} \kappa_6^{(4)} - 8 \kappa_4^{(4)} \kappa_6^{(4)2} + 352 \kappa_6^{(4)3})) + \\ & \kappa_{15}^{(4)2} (41569 \kappa_3^{(4)4} + 1539 \kappa_4^{(4)4} + 32 \kappa_3^{(4)3} (2094 \kappa_4^{(4)} - 505 \kappa_6^{(4)}) + 2106 \kappa_4^{(4)3} \kappa_6^{(4)} - 11745 \kappa_4^{(4)2} \kappa_6^{(4)2} + \\ & 5448 \kappa_4^{(4)} \kappa_6^{(4)3} + 2341 \kappa_6^{(4)4} + 3 \kappa_3^{(4)2} (15579 \kappa_4^{(4)2} - 6648 \kappa_4^{(4)} \kappa_6^{(4)} - 9442 \kappa_6^{(4)2}) + \\ & 2 \kappa_3^{(4)} (7209 \kappa_4^{(4)3} - 972 \kappa_4^{(4)2} \kappa_6^{(4)} - 16128 \kappa_4^{(4)} \kappa_6^{(4)2} + 3560 \kappa_6^{(4)3})))) / \\ & (\kappa_{15}^{(4)} (-3 \kappa_{15}^{(4)} + \kappa_3^{(4)} + \kappa_6^{(4)}) (9 \kappa_{15}^{(4)2} + 2 \kappa_3^{(4)2} + \kappa_4^{(4)} (\kappa_4^{(4)} - 2 \kappa_6^{(4)}) + 2 \kappa_3^{(4)} (\kappa_4^{(4)} - \kappa_6^{(4)}) + \\ & \kappa_{15}^{(4)} (-7 \kappa_3^{(4)} - 6 \kappa_4^{(4)} + 5 \kappa_6^{(4)})))) \end{aligned} $
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	2	0	$ \begin{aligned} & -((-81 \kappa_{15}^{(4)}{}^{(4)}{}^3 + 20 \kappa_3^{(4)}{}^{(4)}{}^3 + 20 \kappa_3^{(4)}{}^{(4)}{}^2 \kappa_4^{(4)}{}^{(4)} + \\ & 10 \kappa_3^{(4)}{}^{(4)}{}^2 \kappa_4^{(4)}{}^{(4)} + 18 \kappa_{15}^{(4)}{}^{(4)}{}^2 (10 \kappa_3^{(4)}{}^{(4)} + 3 \kappa_4^{(4)} - 5 \kappa_6^{(4)}) + 10 \kappa_4^{(4)}{}^2 \kappa_6^{(4)} - 20 \kappa_3^{(4)} \kappa_6^{(4)}{}^2 - 20 \kappa_4^{(4)} \kappa_6^{(4)}{}^2 - \\ & \kappa_{15}^{(4)} (103 \kappa_3^{(4)}{}^2 + 78 \kappa_3^{(4)} \kappa_4^{(4)} + 18 \kappa_4^{(4)}{}^2 - 40 \kappa_3^{(4)} \kappa_6^{(4)} + 6 \kappa_4^{(4)} \kappa_6^{(4)} - 71 \kappa_6^{(4)}{}^2) + \sqrt{(183708 \kappa_{15}^{(4)}{}^6 - \\ & 1458 \kappa_{15}^{(4)}{}^5 (245 \kappa_3^{(4)} + 132 \kappa_4^{(4)} - 37 \kappa_6^{(4)}) + 100 (2 \kappa_3^{(4)}{}^2 + \kappa_4^{(4)} \kappa_4^{(4)} - 2 \kappa_6^{(4)}) + 2 \kappa_3^{(4)} (\kappa_4^{(4)} - \kappa_6^{(4)})^2) (\kappa_3^{(4)} + \kappa_6^{(4)})^2 + \\ & 243 \kappa_{15}^{(4)}{}^4 (1273 \kappa_3^{(4)} + 330 \kappa_4^{(4)}{}^2 + 4 \kappa_3^{(4)} (309 \kappa_4^{(4)} - 97 \kappa_6^{(4)}) - 156 \kappa_4^{(4)} \kappa_6^{(4)} - 89 \kappa_6^{(4)}{}^2) - \\ & 9 \kappa_{15}^{(4)}{}^3 (16558 \kappa_3^{(4)}{}^3 + 1836 \kappa_4^{(4)}{}^3 + 12 \kappa_3^{(4)}{}^2 \kappa_4^{(4)} (1827 \kappa_4^{(4)} - 545 \kappa_6^{(4)}) - 729 \kappa_4^{(4)}{}^2 \kappa_6^{(4)} - \\ & 2700 \kappa_4^{(4)} \kappa_6^{(4)}{}^2 + 1060 \kappa_6^{(4)}{}^3 + 3 \kappa_3^{(4)} (3645 \kappa_4^{(4)}{}^2 - 1800 \kappa_4^{(4)} \kappa_6^{(4)} - 1586 \kappa_6^{(4)}{}^2)) - \\ & 5 \kappa_{15}^{(4)} (\kappa_3^{(4)} + \kappa_6^{(4)}) (1256 \kappa_3^{(4)}{}^4 + 8 \kappa_3^{(4)}{}^3 (289 \kappa_4^{(4)} - 197 \kappa_6^{(4)}) + 2 \kappa_3^{(4)}{}^2 (995 \kappa_4^{(4)}{}^2 - 1400 \kappa_4^{(4)} \kappa_6^{(4)} - 16 \kappa_6^{(4)}{}^2) + 2 \kappa_3^{(4)} \\ & (417 \kappa_4^{(4)}{}^3 - 845 \kappa_4^{(4)}{}^2 \kappa_6^{(4)} + 68 \kappa_4^{(4)} \kappa_6^{(4)}{}^2 + 176 \kappa_6^{(4)}{}^3) + \kappa_4^{(4)} (153 \kappa_4^{(4)}{}^3 - 390 \kappa_4^{(4)} \kappa_6^{(4)} - 8 \kappa_4^{(4)} \kappa_6^{(4)}{}^2 + 352 \kappa_6^{(4)}{}^3)) + \\ & \kappa_{15}^{(4)}{}^2 (41569 \kappa_3^{(4)}{}^4 + 1539 \kappa_4^{(4)}{}^4 + 32 \kappa_3^{(4)}{}^3 (2094 \kappa_4^{(4)} - 505 \kappa_6^{(4)}) + 2106 \kappa_4^{(4)} \kappa_6^{(4)} - 11745 \kappa_4^{(4)} \kappa_6^{(4)}{}^2 + \\ & 5448 \kappa_4^{(4)} \kappa_6^{(4)}{}^3 + 2341 \kappa_6^{(4)}{}^4 + 3 \kappa_3^{(4)}{}^2 (15579 \kappa_4^{(4)}{}^2 - 6648 \kappa_4^{(4)} \kappa_6^{(4)} - 9442 \kappa_6^{(4)}{}^2) + \\ & 2 \kappa_3^{(4)} (7209 \kappa_4^{(4)}{}^3 - 972 \kappa_4^{(4)}{}^2 \kappa_6^{(4)} - 16128 \kappa_4^{(4)} \kappa_6^{(4)}{}^2 + 3560 \kappa_6^{(4)}{}^3)))/ \\ & (\kappa_{15}^{(4)} (-3 \kappa_{15}^{(4)} + \kappa_3^{(4)} + \kappa_6^{(4)}) (9 \kappa_{15}^{(4)}{}^2 + 2 \kappa_3^{(4)} + \kappa_4^{(4)} (\kappa_4^{(4)} - 2 \kappa_6^{(4)}) + 2 \kappa_3^{(4)} (\kappa_4^{(4)} - \kappa_6^{(4)}) + \\ & \kappa_{15}^{(4)} (-7 \kappa_3^{(4)} - 6 \kappa_4^{(4)} + 5 \kappa_6^{(4)}))) \end{aligned} $
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	1	$-\frac{3 \kappa_3^{(2)}}{3 \kappa_1^{(4)} - 3 \kappa_{15}^{(4)} + 3 \kappa_2^{(4)} + \kappa_3^{(4)} + \kappa_6^{(4)}}$	$-\frac{2}{3 \kappa_1^{(4)} - 3 \kappa_{15}^{(4)} + 3 \kappa_2^{(4)} + \kappa_3^{(4)} + \kappa_6^{(4)}}$
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	3	$\frac{2^{(2)}_{\kappa_3}}{3^{(4)}_{\kappa_{15}} 2^{(4)}_{\kappa_2}}$	$-\frac{4}{9^{(4)}_{\kappa_{15}} 6^{(4)}_{\kappa_2}}$
Resolved unitarity condition(s):	(Demonstrably impossible)		